

Draining Patience or Fueling Creativity? The LLM Effect on KAIST students

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1. Introduction

Large Language Models (LLMs) have rapidly become powerful tools for augmenting human performance in a variety of language-based tasks. Recent advances have expanded their application from simple translation and text generation to more complex domains such as coding, mathematical problem solving, scientific reasoning, academic writing, and idea generation. The latest LLMs demonstrate the capacity for contextual reasoning, generating not only draft solutions but also step-by-step guidance and meaningful alternatives. Field experiments have shown that access to LLMs can improve both throughput and output quality in real-world tasks (Brynjolfsson et al., 2023), while lab-based studies confirm that coding assistants increase task efficiency and user satisfaction (Birhane et al., 2022). These results support the view that LLMs can serve as intelligent knowledge work assistants beyond simple automation.

However, despite these productivity gains, growing concerns have emerged regarding the broader cognitive implications of LLM use. Most prior studies have focused on outcome-level performance metrics or product quality, while empirical evidence on how LLMs reshape **human cognitive behaviors**—particularly those involved in the process of problem solving—remains limited. Of special importance are three high-level cognitive functions: **patience**, **concentration**, and **creativity**. These faculties are not only crucial for effective learning and academic achievement but also fundamental to long-term human adaptability and decision-making.

First, **patience**, or the ability to persist in the face of difficulty, is vital for tackling unfamiliar or complex problems. While LLMs provide immediate access to potentially correct or helpful outputs, this convenience may diminish users' motivation to persist through uncertainty. Studies on cognitive offloading suggest that reliance on external tools reduces the perceived need for internal problem-solving effort, ultimately lowering persistence (Risko & Gilbert, 2016). Over time, such reliance can erode cognitive resilience, especially in educational settings, where productive struggle is a known precursor to deep learning (Duckworth et al., 2007). In contrast, repeated success through independent effort enhances self-efficacy and long-term motivation—outcomes that are harder to achieve when users default to AI support.

Second, **concentration**, or the ability to sustain attention over time, may also be altered by the presence of LLMs. The speed and fluency of AI-generated answers can shorten the user's engagement with the full reasoning process, reducing the cognitive investment required to maintain task focus. This is particularly problematic in tasks that demand continuous attention, such as mathematical modeling or scientific interpretation. Research on attention in digital environments shows that easy access to external information correlates with decreased mental persistence and increased susceptibility to distraction (Uncapher & Wagner, 2018; Mark et al., 2008). As a result, LLMs may unintentionally undermine users' ability to sustain deep cognitive engagement over extended periods.

Third, **creativity**—the capacity to generate, explore, and refine novel ideas—presents a particularly nuanced case. On one hand, LLMs can serve as effective idea generators, offering a broad range of possibilities that may spark creative thinking. On the other hand, over-reliance on these suggestions can reduce user agency, turning idea generation into a process of selecting rather than creating. For

example, users may find themselves choosing among pre-generated AI responses rather than constructing original solutions from scratch. Recent studies indicate that LLM usage can lead to stylistic and structural convergence in user-generated texts, as users tend to adopt model-suggested phrasings and formulations, potentially reducing individual variability and originality (Lin & Zhu, 2024; Briesch et al., 2023).

Crucially, the cognitive effects of LLM usage are likely to vary significantly by **task domain**. Tasks such as mathematics or science tend to be answer-oriented and logic-driven, where structured support from LLMs may enhance learning or accuracy. Conversely, tasks like writing or ideation are more open-ended, requiring subjective interpretation, audience awareness, and personal voice—all of which may be more vulnerable to the constraining effects of LLM assistance. Despite this likely variation, most previous studies have examined only a single domain, often through self-reported impressions rather than behavioral data. There is a clear gap in our understanding of how LLMs impact patience, concentration, and creativity across different domains of human cognition.

To address this gap, the present study investigates the effects of LLM usage across five task domains—coding, writing, mathematics, science, and ideation—using a within-subjects design involving KAIST undergraduate students. Participants complete tasks with and without the support of a LLM, allowing for direct, within-person comparison across conditions and domains. The following section outlines the specific research questions and objectives of this study in detail.

2. Problem Statement & Objectives

2.1 Problem Statement

While LLMs are proven to enhance productivity, empirical evidence on their cognitive impact on the human problem-solving process is limited. This study focuses on three key cognitive functions:

- **Patience:** The immediate answers from LLMs may diminish users' motivation to persist through uncertainty.
- **Concentration:** The speed of AI answers may shorten deep cognitive engagement.
- **Creativity:** Reliance on AI suggestions may turn the process from "creating" into "selecting".

Crucially, these cognitive effects likely vary by task domain, yet existing research is mostly limited to a single domain or relies on self-reports, not behavioral data. A "clear gap exists" in our understanding of the impact across different domains based on behavioral data.

2.2 Objectives

To address this gap, this study investigates the effects of LLM usage across five task domains (coding, writing, mathematics, science, and ideation) using a within-subjects design.

The specific objectives are:

- **To measure cognitive functions:** To empirically measure the main effect of LLM usage on **patience**, **concentration**, and **creativity** using objective behavioral indicators.
- **To analyze cross-domain effects:** To statistically test for an Interaction Effect to see if the LLM's impact differs across the five task domains.
- **To study the LLM as a 'Collaborative Partner':** To use tasks with 'Core Barriers' to ensure participants engage in higher-order cognitive activities and position the LLM as a 'Collaborative Partner', not just an answer generator.

3. Research Methods

3.1 Experiment Participants

The participants of this study will consist of **50 undergraduate students from KAIST**. As this study adopts a **within-subjects design**, where all participants experience all experimental conditions, it is expected that a sufficient amount of data will be secured from the 50 participants, providing adequate statistical power to explore the relationships between variables.

3.2 Experiment Design

3.2.1 Variables

- **Independent Variables: Task Condition** (A condition with the use of an LLM vs. a condition without the use of an LLM), and **Task Domain** (Coding, Writing (Reports/Translation), Mathematics, Science, and Ideation).
- **Dependent Variables: Patience, Concentration, and Creativity** (measured as the change score between pre-task and post-task measurements).
- **Control Variables:** Experimental environment (location, time), and the type of LLM used (unified as **Google's Gemini 2.5 Pro**).

3.2.2 Design Method

This study employs a **2 (Task Condition) x 5 (Task Domain) within-subjects factorial design**.

Core Design Strategy: For each task domain, two problems of equivalent difficulty (A and B) will be developed in advance. Participants will solve each of the two problems under different conditions (with/without LLM). The assignment of which problem (A or B) is solved under which condition will be **counterbalanced** across participants to offset any confounding effects arising from the combination of a specific problem with a specific condition.

3.2.3 Task Design Principles

The tasks used in this study are designed to position the Large Language Model (LLM) not as a mere 'answer generator' but as a 'Collaborative Partner' to the human participant. To achieve this, all tasks intentionally include 'Core Barriers' that prevent the LLM from solving them independently.

- **Information Asymmetry:** Requires the human to provide local information that the LLM has not been pre-trained on (e.g., requiring code to be written based on a fictitious KAIST API specification).
- **Multi-step Reasoning:** Requires human strategic judgment and step-by-step instructions for the problem-solving process (e.g., requiring the participant to first select the most appropriate mathematical model for given data, justify the choice, and then solve it).
- **Open-ended Evaluation:** Involves tasks without a single correct answer, requiring the human to evaluate and refine multiple alternatives generated by the LLM (e.g., requiring the participant to write a balanced column for a specific audience by referencing two opposing research papers).

This design ensures that participants engage in higher-order cognitive activities such as strategizing, providing information, and critically evaluating results, rather than simply copying the LLM's output.

3.3 Experiment Procedure

1. **Preparation:** Develop pairs of problems with equivalent difficulty for each of the five domains (10 problems total) and validate their equivalence through a pilot test.
2. **Pre-screening:** Survey prospective participants to ensure they possess the minimum required knowledge for the five domains, and recruit only qualified individuals.
3. **Experiment Execution:** Participants will complete sessions for all five task domains in a randomized order (10 tasks total). Dependent variables will be measured immediately before

and after each task. The order of task domains and conditions will be randomized for each participant.

3.4 Measurement Tools

This study will measure the dependent variables using the following tools, focusing on objective and sensitive behavioral indicators.

- **Patience: Behavioral Measurement using an 'Unsolvable Task'**
 - **Description:** Participants are presented with a problem that appears ordinary but is intentionally designed to be unsolvable (e.g., a puzzle with a hidden logical contradiction). Participants are instructed to press a 'give up' button when they recognize their limit.
 - **Metric:** The **persistence time (in seconds)** from the start of the task until the participant presses the 'give up' button. A longer duration is interpreted as higher frustration tolerance, or patience.
- **Concentration: SART (Sustained Attention to Response Task)**
 - **Description:** Digits from 1 to 9 appear on the screen at 1.5-second intervals. Participants must press the spacebar for all digits except for the number '3'. The task lasts approximately 5 minutes.
 - **Metrics:** ① **Commission Errors:** The number of times the spacebar is incorrectly pressed for the number '3'. ② **Mean Response Time:** The average speed (in ms) of correct responses to other digits. An increase in errors or a slower response time is interpreted as a decline in concentration.
 - **Reference:** Templates for SART can be easily found on online psychology experiment platforms such as PsyToolkit.
- **Creativity: AUT (Alternative Uses Test)**
 - **Description:** Participants are given a common object (e.g., "a brick," "a paper clip") and are asked to list as many creative and novel uses as possible within a two-minute time limit. Different objects are used for pre- and post-task measurements (e.g., 1st - paper clip, 2nd - brick) to prevent learning effects.
 - **Metric:** ① **Fluency:** The total number of ideas. A higher number is interpreted as more active divergent thinking. ② **Originality:** The statistical rarity of the ideas (Score ideas based on how few other participants generated them).

3.5 Data Analysis Plan

1. **Data Preprocessing:** The scores of the dependent variables measured before and after each task will be converted into a '**Change Score**' (Post-task score - Pre-task score).
2. **Primary Analysis Method:** The collected data will be analyzed using a **Two-Way Repeated Measures ANOVA**.
3. **Hypothesis Testing:** This analysis will be used to test for the **Main Effect** of LLM usage and the **Interaction Effect**, which examines whether the effect of the LLM differs across the various task domains.

4. Expected Results & Significance

4.1 Expected Results

Main Effects: We expect LLM use will decrease patience and concentration. For creativity, we predict a '**convergence effect**' : while **fluency** may increase, we hypothesize that **originality** will significantly decrease.

Interaction Effects: We hypothesize these negative impacts will be amplified when users face the 'Core Barriers' (where the LLM fails), as this will induce frustration. This effect is expected to be more pronounced in open-ended domains like **Writing** and **Ideation**.

4.2 Significance

Academic Contribution: Provides crucial **behavioral** and **cross-domain** data, addressing a "clear gap" in current research.

Educational Implications: Informs guidelines for educators on how to use AI while preserving 'productive struggle' and cognitive resilience.

Methodological Contribution: The 'Core Barrier' task design offers a new paradigm for studying genuine human-AI **collaboration**.

5. Timeline

This project will take 6 weeks, from Week 9 to Week 14. The final week will be for the report and presentation preparation.

Week 9 (Preparation & Pilot Test): Finalize the 10 experiment problems (5 domains x 2 versions). Run a pilot test to check problem difficulty and timing. Prepare the measurement tools (SART, AUT templates).

Week 10 (Recruit Participants & Start): Recruit 50 KAIST undergraduate students. Screen participants for required knowledge. Begin the main experiment sessions.

Week 11-12 (Main Experiment): Complete the experiment with all 50 participants. Focus on collecting all pre/post measurement data for all 10 tasks.

Week 13 (Data Analysis): Process the collected data. Run statistical analysis to test hypotheses and find key results.

Week 14 (Final Report & Presentation Preparation): Interpret the findings from the analysis. Write the complete final report. Prepare the final presentation slides.

6. Reference

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